

Camp Overview

This is a full plan for a **Arduino Robotics & Electronics** camp of **51 children** on Saturday 15 August 2026. The intake is split into age bands — 7-9 yrs (20), 10-12 yrs (19), 13-15 yrs (12) — and each band into small groups so every child works at the right level and gets plenty of hands-on time. The sections below give the group split, staffing, the master timetable, and a tailored programme for each band.

Groups at a Glance

Each group stays together all day with its own lead facilitator, so no child is ever in a group larger than the target size.

AGE BAND	GROUP	SIZE
7-9 yrs	7-9 yrs · Group A	10 children
7-9 yrs	7-9 yrs · Group B	10 children
10-12 yrs	10-12 yrs · Group A	10 children
10-12 yrs	10-12 yrs · Group B	9 children
13-15 yrs	13-15 yrs · Group A	6 children
13-15 yrs	13-15 yrs · Group B	6 children

Staffing & Ratios

With **51 children** in **6 groups**, this plan needs **10 adults** on the day — an overall ratio of about **1 adult to 5.1 children** (target 1:10 or better).

ROLE	PEOPLE
Lead facilitators (one per group)	6

ROLE	PEOPLE
Floating helpers	2
Camp coordinator	1
First-aider	1
Total adults	10

Master Day Timetable

The whole camp runs on one clock. Groups start and stop together so breaks, lunch and the showcase stay in sync.

TIME	ACTIVITY
08:30 – 09:00	Arrival, registration & name badges
09:00 – 09:20	Opening assembly — welcome, safety brief, split into groups (all bands together)
09:20 – 10:45	Project Block 1 — each group in its own space
10:45 – 11:05	Morning break & snack (staggered so the space doesn't crowd)
11:05 – 12:30	Project Block 2 — build continues / next activity
12:30 – 13:15	Lunch & free play (supervised, outdoors if possible)
13:15 – 14:30	Project Block 3 — finish builds, test & debug
14:30 – 14:50	Showcase prep — groups tidy kits and rehearse a 60-second demo
14:50 – 15:20	Showcase & demos — every group presents to the whole camp
15:20 – 15:30	Closing circle, certificates & departure

Programme by Age Band

7-9 yrs (2 groups: 7A, 7B)

- **Block 1:** Button Controlled LED — push button toggles an LED on/off.
- **Block 2:** Traffic Light Simulation — three LEDs sequence like a real traffic light.
- **Block 3:** RGB LED Colour Mixer — three potentiometers blend red, green & blue.

Learning goals:

- Understand a simple circuit: power, resistor, LED, ground.
- See how code statements run in order and repeat (loops, delays).
- Discover the difference between an on/off (digital) signal and a variable (analog) one.

Facilitator brief: Pre-build one working sample of each project so children can compare their circuit against a "gold standard." Keep instructions to one step at a time with a visual check-in before moving on — this age group loses focus if given the full sequence at once. Watch for reversed LED legs (flat edge = short leg = ground) and loose breadboard jumpers, the two most common faults. Have a helper roam continuously rather than teach from the front. Celebrate every blink loudly — confidence matters more than speed here.

10-12 yrs (2 groups: 10A, 10B)

- **Block 1:** Morse Code Blinker — an LED blinks out a message in Morse code.
- **Block 2:** Ultrasonic Distance Sensor — an HC-SR04 measures distance and an LED warns on proximity.
- **Block 3:** Servo Motor Control — a potentiometer steers a servo to any angle.

Learning goals:

- Translate a real-world code (Morse) into timed digital output.
- Use sensor input and conditional logic (if/else, thresholds) to trigger an output.
- Map analog input (potentiometer) to actuator output (servo angle) using `map()`.

Facilitator brief: This band can read wiring diagrams themselves — supply printed diagrams rather than building for them. Pre-flash a test sketch onto each Arduino at the start of the day to confirm boards work before kids start wiring. For the ultrasonic block, check sensor orientation (eyes face outward) and remind them not to touch the sensor face while testing. For servos, warn about stall current if the horn is physically blocked — disconnect power before adjusting the horn position. Encourage pairs to debug together before calling a facilitator ("ask three then me").

13-15 yrs (2 groups: 13A, 13B)

- **Block 1:** Fan Control — a temperature sensor drives a relay-powered fan.

- **Block 2 & 3 (continuous project):** Simon Says Game — classic memory game with LEDs, buttons and buzzer. Block 2 = wiring + core logic; Block 3 = debugging, stretch features (scoring, increasing speed, sound effects) and polish.

Learning goals:

- Understand relay switching to control higher-power loads safely from a microcontroller.
- Design and implement a simple state machine / game logic in code.
- Practise independent iterative debugging: read error messages, isolate faults, test incrementally.

Facilitator brief: Treat this band as junior engineers — give them the goal and datasheet-style pinouts, not step-by-step instructions. For the fan control block, brief them explicitly on relay safety (never touch relay contacts while powered, keep mains-side wiring away from the low-voltage side even though this build uses a 5V fan). Since Simon Says spans two blocks, help each group write a short task list at the start of Block 2 so they can self-track progress across the lunch break. Reserve time in Block 3 purely for debugging — don't let them start new features after 14:00. Have stretch-goal cards ready (add a "strict mode," add a high-score EEPROM save) for groups that finish early.

Materials Master List

ITEM	SPEC	PER GROUP	GROUPS	TOTAL NEEDED
Arduino Uno R3 (or compatible)	ATmega328P board	1	6	6
USB cable	Type A-B, 60cm	1	6	6
Breadboard	830 tie-point, solderless	1	6	6
Jumper wire pack	M-M / M-F / F-F assorted, ~65pcs	1	6	6
Resistor pack	Mixed 220Ω & 10kΩ, 10pcs	1	6	6
LED 5mm	Red	2	2 (7-9)	4
LED 5mm	Yellow	1	2 (7-9)	2

LED 5mm	Green	1	2 (7-9)	2
Push button switch	6mm tactile	1	2 (7-9)	2
Potentiometer	10kΩ linear rotary	3	2 (7-9)	6
RGB LED module	Common-cathode, 5mm	1	2 (7-9)	2
LED 5mm	Red (Morse blinker)	1	2 (10-12)	2
Ultrasonic sensor	HC-SR04	1	2 (10-12)	2
Servo motor	SG90 micro servo	1	2 (10-12)	2
Potentiometer	10kΩ linear rotary (servo control)	1	2 (10-12)	2
Temperature sensor	DHT11 module	1	2 (13-15)	2
Relay module	5V single-channel	1	2 (13-15)	2
DC fan	Small 5V brushed fan	1	2 (13-15)	2
Push button switch	6mm tactile (Simon Says)	4	2 (13-15)	8
LED 5mm	Assorted red/green/blue/yellow (Simon Says)	4	2 (13-15)	8
Piezo buzzer	5V active buzzer	1	2 (13-15)	2
Laptop with Arduino IDE installed	Windows/Mac, USB-A port	1	6	6
Power strip / extension	4-way, 5m cable	1	3 (shared across pairs of stations)	3

Shared kit note: Power strips are shared between two neighbouring stations rather than issued per group. Keep 2 spare Arduino Uno boards, 2 spare USB cables and one spare of each sensor/actuator (HC-SR04, servo, DHT11, relay) at the facilitator table as swap-out stock — these are not counted in the totals above. Nerokas or Pixel Electronics can supply all listed items locally; order a 10% buffer on small parts (LEDs, resistors, buttons) to cover breakages.

Safety, Logistics & Contingencies

Room/space plan: Six clearly marked stations (one per group) arranged around the main hall or in two adjoining rooms, each with its own table, kit tray and

facilitator. 7–9 groups closest to the supervising coordinator's line of sight; 13–15 groups given the most physical space to spread out wiring diagrams and stretch-goal notes.

Movement between blocks: Kits and half-finished builds stay at each group's station throughout the day — only the children move (to breaks, lunch, showcase prep). This avoids disturbing wiring and lets facilitators reset quickly between blocks.

Allergy / first-aid notes: Confirm allergy list against registration forms before snack is served; keep nut-free and dairy-free snack options clearly labelled and separated. Designated first-aider present at all times with a stocked kit visible at the facilitator table; any epi-pens/medication for named children stored at that table, not in bags. No bare wires near mouths — remind younger groups not to touch battery terminals or connectors together.

Wet-weather contingency: Move lunch/free play indoors into the hall; stagger by age band (youngest first) to manage space. Have quiet indoor games (card games, simple puzzles) ready as an alternative to outdoor play.

Early-finisher contingency: Each band has a stretch card ready — 7–9 add a fourth LED colour pattern to the traffic light; 10–12 add a buzzer alert to the ultrasonic sensor; 13–15 add scoring/EEPROM save to Simon Says. Early finishers may also help a struggling peer within their own group before facilitators are called.

Equipment-failure contingency: Spare Arduino boards, USB cables and one spare of each specialised sensor are kept at the facilitator table (see Materials note). Facilitators use a labelled swap procedure: unplug, tag faulty unit "FAULTY – DO NOT USE", replace, continue. If a laptop fails, groups can share the neighbouring station's laptop in short rotations while it charges/restarts.

Staff Briefing Checklist

- Confirm all 6 stations are set up with correct per-band kit before 08:30 doors-open.
- Flash a known-working test sketch to every Arduino board to confirm it powers up and uploads correctly.

- Check allergy/medical list against today's registration and snack ingredients.
- Confirm first-aider identity and kit location; brief all facilitators on it.
- Walk through each band's three-block project sequence with the relevant lead facilitator; confirm they have the sample "gold standard" build where required (7-9 band).
- Distribute stretch-goal cards for early finishers to each station.
- Confirm spare components (boards, cables, sensors) are stocked and clearly labelled at the facilitator table.
- Agree the swap-out procedure for faulty equipment with all facilitators.
- Confirm wet-weather indoor plan and who moves furniture/kit if needed.
- Assign one facilitator per age band to coordinate the 60-second showcase demo rehearsal.
- Confirm break/lunch staggering times per group to avoid overcrowding.
- Check power strips and extension leads are safely routed (no trailing cables across walkways).
- Brief all staff on emergency exits, assembly point and closing-circle logistics.